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# Recent advances in the mechanical characterization of shales at nano-to micro-scales: A review

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ABSTRACT

Understanding geomechanical properties of shales, such as stiffness properties and fracture toughness, is important in different areas of petroleum industry-related activities. Nanoscale to microscale experiments on shale samples require a much smaller sample compared to macroscale experiments. This is a major advantage because irregularly-shaped drill cuttings from shales can be used for experimental analyses, such as nano-indentation or atomic force microscopy (AFM). However, characterization of mechanical properties at the nano- to micro-scale is a relatively new addition to classical geomechanical experiments on shales, and there is still a lack of both fundamental knowledge and standard procedures for conducting experiments at this scale. The theoretical principles of nanoindentation-based (Gridded Nanoindentation and Modulus Mapping) and AFM-based experimental methods (PeakForce QNM™ and AFM-IR), a review of the literature results and the major findings are explained in Sections 2 and 3, respectively. These experimental techniques are compared in Section 4. Overall, nanoindentation experiments reveal the anisotropy of organic matter (OM)-rich shales at the nano-scale, and it is shown that gridded nanoindentation is capable of characterizing different minerals with respect to their stiffness properties. The capability of AFM (PeakForce QNM™) to differentiate between different minerals within OM-rich shales with distinct mechanical properties, in particular stiffness, is also demonstrated. It is shown that AFM-based methods are more straightforward than gridded-nanoindentation to characterize OM-rich shales based on the stiffness of different constituent minerals of OM-rich shales, but nanoindentation can apply higher loads than AFM-based methods. Therefore, they can measure creep properties and fracture toughness of OM-rich shales. It is shown that the OM is the most compliant part of the OM-rich shales, but it is still unclear if thermal maturation has an impact on the stiffness of OM. It is necessary to investigate if there is a relationship between the OM type (kerogen, bitumen, etc.) and mechanical properties such as Young's modulus and Creep rate.

## 1. Introduction

Shales (or mudrocks) are the source rock for hydrocarbons in "shale play reservoirs" and are also the caprock in most conventional reservoirs. Shales provide a sealing capability for geological and anthropological waste repositories, such as carbon dioxide sequestration and nuclear waste containment. One of the difficulties facing petroleum mudrock reservoir development has been the accurate assessment of the host rock's mechanical properties. These properties influence vital reservoir aspects such as borehole stability, fracturing behavior, yield strength and elastic moduli (Dewhurst et al., 2011; Fam et al., 2003), affecting not only fracturing efficiency, but also the petroleum flow to the wellbores. Consequently, an adequate knowledge of the mechanical properties of shales is crucial to the sustainable development of unconventional reservoirs.

Shales are among the most complex geological materials, as their composition varies through both time and space and at different scales, from nano-to macro-scopic (Ilgen et al., 2017). This heterogeneity leads to the anisotropy of properties and a range of macroscopic behaviors. Individual clay grains exist at the nanoscale and are the building block of shales (Grim, 1988; Mitchell and Soga, 2005). Although the structures of common clay minerals, such as kaolinite and montmorillonite, are well established, their small sizes make it difficult to directly study their mechanical properties. At the sub-micron level, advanced observational methods such as Scanning Electron Microscopy (SEM) and Transmission Electron Microscopy (TEM) can be applied to study the geochemical

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properties of clay mineral composites (Bennett et al., 1991; Mitchell and Soga, 2005). Using Small Angle Neutron Scattering (SANS) (Hall et al., 1983) and Mercury Intrusion Porosimetry (MIP) (Hu et al., 2017), it has been shown that the spaces between clay sheets represent the major portion of the total porosity of shale, and its characteristic pore access radius is in the range of several nanometers. In the petroleum industry, most analytical methods are focused on the macroscale (sub-millimeter to millimeter). This level has classically been designated as the scale of shale acoustics and exploration geophysics-related research into shear and compressional wave velocities (Dewhurst and Siggins, 2006; Domnesteanu et al., 2002; Jakobsen and Johansen, 2000; Sayers, 1999; Wang, 2002).

Traditionally, mechanical characterization of rock has been conducted using uniaxial or triaxial compression tests (Kumar et al., 2017), and laboratory tests or logging tools provide sonic and density data with which to obtain the elastic properties, including Young's modulus and Poisson's ratio (Eissa and Kazi, 1988; Sone and Zoback, 2013). Because of the existing instability of chemical and mechanical properties of shale formations, obtaining a core stable enough to perform traditional geomechanical experiments can be difficult and costly (Liu et al., 2016). However, progress in nanomechanics and nanotechnology has made it possible to investigate the local mechanical properties of shales at the nano-to micro-scale by nanoindentation using a small sample volume (Liu et al., 2016). Nanoindentation was developed in 1970s to evaluate the hardness of a small volume of material. In nanoindentation, the loads and displacements of an indenter are continuously controlled and monitored when the indenter is driven into, and withdrawn from, a small sample volume.

The results from the published work on mechanical characterization of OM-rich shales at the nanoscale show that there is a significance difference between the modulus of different components of shales, and the OM is the softest mechanical phase (Eliyahu et al., 2015; Eichmann et al., 2018; and Slim et al., 2019). Molecular simulations at the nanoscale were implemented to understand why OM is the softest mechanical phase and how the mechanical properties of OM change under different conditions, such as i) increasing thermal maturity (Bousige et al., 2016; Kashinath et al., 2020), or ii) in different typical macroscale geomechanical experiments (Wu and Firoozabadi, 2020a), and iii) change in OM type and maturity level on the tensile behavior of OM (Wu and Firoozabadi, 2020b). Most of the published research on mechanical characterization of OM-rich shales at the nanoscale and microscale did not identify how they identified type of OM (bitumen, kerogen, etc.) in their papers; therefore, the term of OM is used for those papers. If the published research explicitly mentioned the type of OM, the method and type of OM is reported in this review paper.

There has been some recently published research work on upscaling the measured mechanical properties of OM-rich shale from nano- and micro-scales to the macro-scale. Goodarzi et al. (2016) described all the homogenization methods that are applicable to obtaining a mechanical response of composite materials, and they investigated their accuracy in predicting the elastic response of OM-rich shales. Abedi et al. (2016) focused on understanding the impact of change in thermal maturity and total organic content (TOC) on the effective mechanical properties of the clay/OM phase. Monfared and Ulm (2016) & Abedi et al. (2016) found that the thermally mature OM-rich shales have a granular morphology in the microscale to millimeter scale. Mashhadian et al. (2018) applied the Abedi et al. (2016) model to upscale the experimental measurements of elastic properties of carbonate-rich shale samples, and they quantified the uncertainty of model input parameters (porosity, indentation modulus and hardness, bulk modulus and shear modulus) to optimize the reliability and robustness in predicting the homogenized poroelastic properties. Their samples were outcrop OM-rich and OM-free Eagle Ford shales. Dubey et al. (2019) used the same framework of Mashhadian et al. (2018) and incorporated the existing microfractures in OM-rich shales. Liu et al. (2018c) used the Mori-Tanaka (MT) homogenization method (one of the methods explained in Goodarzi et al., 2016) to upscale the mechanical properties (Young's modulus) measured with the gridded-nanoindentation method. The homogenization methods applied to synthetic composite material to consider the microstructural arrangements, material properties and volume fraction of defined constituents such as matrix and inclusions within the matrix. Moreover, they conducted traditional unconfined compressional tests on core samples of OM-rich shales. The upscaled results from the MT scheme were compared with the measured mechanical properties (Young's modulus) from the unconfined compressional test. This comparison showed that the difference between the results of the homogenization method and traditional unconfined compressional tests is less than 14.5%, which is an acceptable difference.

Given recent advancements in measurements of the mechanical properties of OM-rich shales from the nanoscale to microscale, and the lack of detailed review on these advancements, we compile and review here relevant studies to identify non-traditional methods for the mechanical characterization of shale at these very small scales. First the nanoindentation-related and AFM-based methods are described and their applications are explained, then their advantages and limitations are compared.

## 2. Nanoindentation

The nanoindentation method is used to measure mechanical properties such as stiffness and hardness of materials at the nano-to microscale. Nanoindentation has been used in material science research on various materials such as fibers, thin films, concrete, and bones (Ago et al., 2013; Constantinides et al., 2006; Schwiedrzik et al., 2014).

### 2.1. Methods of applying nanoindentation

Gridded nanoindentation and modulus mapping are the two main methods used to investigate the mechanical properties of shales.

#### 2.1.1. Gridded nanoindentation

In general, the three main stages of a gridded indentation test include: (1) a compressive loading stage in which the load reaches a maximum level of $P_{max}$; (2) a holding stage where the load is kept constant throughout; and (3) a continuous unloading stage. Fig. 1 shows a schematic of a load-displacement ($P$-$h$) curve for a typical nanoindentation test. The mechanical properties, including indentation hardness $H$ and the indentation modulus $M$, are derived from the continuum scale mechanical model (Constantinides and Ulm, 2007):

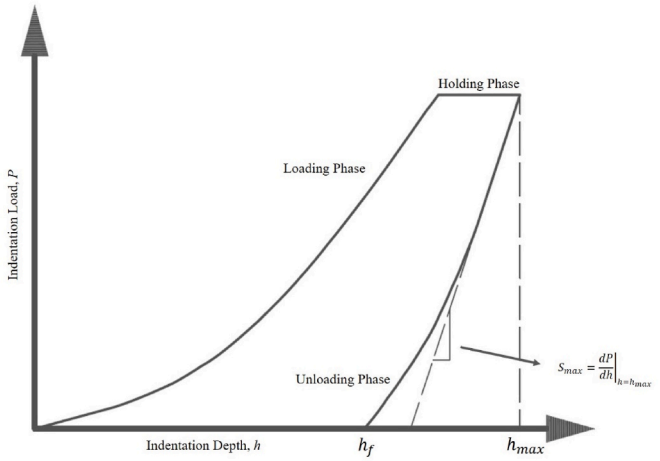



**Fig. 1.** A load-displacement ($P$-$h$) curve for a typical nanoindentation test; $h_f$ is the residual displacement recorded after the unloading stage, and $P_{max}$, and $h_{max}$ are the maximum load and associated displacement, respectively.

$$H \overset{def}{=} \frac{P}{A_c} \tag{1}$$

$$M \overset{def}{=} \frac{\sqrt{\pi}}{2} \frac{S}{\sqrt{A_c}} \tag{2}$$

where all of the required quantities, except for the projected area of contact $A_c$, for determining $H$ and $M$ are directly obtained from the recorded $P$-$h$ curves for each experiment. $P_{max}$, $h_{max}$ are the maximum applied force and maximum indentation depth, respectively. The contact stiffness (or unloading indentation stiffness) is defined as the slope of the upper part of the unloading curve $\left(S = \left(dP/dh\right)_{h=h_{max}}\right)$, and $h_f$ is the residual indentation depth after full unloading.

According to Ulm and Abousleiman (2006), "gridded nanoindentation is useful to characterize the mechanical properties of natural composite materials." The technique includes the assumption that the composite material subject to mechanical characterization is composed of two group of materials with a characteristic length of $D$ and distinguishable mechanical properties. A single indentation test can characterize the mechanical properties of either constituents of the composite material, providing that the characteristic length of the phases ($D$) is much larger than the indentation depth ($h$), $D \gg h$. Other important conditions include: a large number of tests ($N \gg 1$) are conducted based on a grid pattern whose grid spacing ($l$) is larger than the indentation influenced length. Consequently, two neighboring indentations do not interfere with each other. Moreover, this method requires that the grid spacing be much larger than the characteristic length of the two composing materials ($l\sqrt{N} \gg D$). Therefore, the location of each individual indentation test is not statistically biased regarding the spatial distribution of the two composing materials. These conditions for indentation depth, the number of tests, and the length of grid spacing result in the probability of having contact between the indenter tip and one or the other composing materials being identical to the surface fraction occupied by the related composing material on the gridded-nanoindentation surface.

Geological materials are complex and have more than two materials (or minerals) in their structure. To analyze their composite behavior, it is necessary to apply statistical methods, mainly a deconvolution technique, to construct the histogram of each mechanical property over the study area (nanoindentation area). For more information on the deconvolution analysis, readers are referred to Ulm and Abousleiman (2006) & Constantinides and Ulm (2007).

#### 2.1.2. Modulus mapping

Modulus mapping is another approach to nanomechanical characterization. It provides the change of modulus over a specific area comprising different materials. Some papers refer to this method as nano-DMA (Liu et al., 2018a). In this method, a dynamic force is applied to the indenter as it scans the sample surface, and it oscillates periodically. A deformation occurs during the scan procedure. Both force and deformation oscillate with the same frequency. The modulus in the dynamic test is a complex modulus with two components including storage (elastic) and loss (damping) moduli. For more detailed information, readers are referred to Herbert et al. (2008). Subsequently, the storage stiffness of the sample, $K'$, is driven by the equation given in Syed Asif et al. (2001):

$$K' = \frac{F_0}{X}\cos(\phi) + m\omega^2 - K_i \tag{3}$$

where $X$ is the displacement amplitude of the indenter, $F_0$ is the amplitude of applied dynamic force, $\omega$ is the oscillation frequency of the dynamic force and deformation, $m$ is the mass of the indenter tip, $\phi$ is the phase angle that represents the phase difference between the dynamic force and resulting deformation, and $K_i$ is the stiffness of the nanoindenter. Assuming that there exists a spherical contact between the indenter tip and the sample surface, the contact area is calculated as a circular area whose radius is equal to the contact radius $a$ (Johnson, 1985):

$$a = \left(\frac{3F_0R}{4K'}\right)^{\frac{1}{2}} \tag{4}$$

where $R$ is the tip radius. The reduced storage modulus and Young's modulus are different, although they are related. Young's modulus is defined as the tensile elastic modulus, and it is measured in a linear elastic material. The reduced storage modulus $E'$ is derived from the following equation (Oliver and Pharr, 1992; Syed Asif et al., 2001):

$$E' = \frac{K'\sqrt{\pi}}{2\sqrt{A}} \tag{5}$$

where $A$ is the projected contact area, and its radius is calculated by Eq. (4). Moreover, the measured moduli using dynamic modulus mapping or gridded nanoindentation intrinsically contain the elastic properties of the sample and the indenter tip. These moduli are reported as "reduced" moduli. Based on the work of Syed Asif et al. (2001), $E'$ is defined as the reduced elastic modulus for materials that do not display damping, and the reduced elastic modulus, $E_r$, is derived from the following equation:

$$\frac{1}{E_r} = \frac{1-\nu_s^2}{E_s} + \frac{1-\nu_i^2}{E_i} \tag{6}$$

where, $E$ and $\nu$ are Young's modulus and Poisson's ratio. The subscripts of $i$ and $s$ are related to the indenter tip and sample, respectively.

### 2.2. Gridded nanoindentation application

Table 1a represents the experimental properties of nanoindentation testing (gridded-nanoindentation and modulus mapping) applications by different researchers to investigate the mechanical behavior of OM-rich shales. Limitations in their experimental techniques are mentioned; Table 1b is just the continuation of Table 1a. Ulm and Abousleiman (2006) investigated a possible connection between the mineralogy of clay particles and their mechanical properties. Shale specimens were cored in three perpendicular directions. The sample's roughness was measured using AFM (described in Section 3), and the mineral compositions of samples were determined using X-ray diffraction (XRD) measurements. The authors calculated the clay packing density ($\eta$) by measuring the total porosity ($\phi$) of the shale cuttings (Eq. (11) in Ulm and Abousleiman, 2006). The authors then compared the results from gridded nanoindentation with the results of the dynamic ultra-pulse-velocity (UPV) measurements of the overall composite stiffness of shales.

Comparison between results from gridded nanoindentation and UPV measurements showed four important reasons for anisotropy of shales at the macroscopic scale (Ulm and Abousleiman, 2006): (1) One order-of-magnitude difference between the previously measured stiffness values of clay minerals and the measured elasticity content of the clay fabric at the scale of hundreds of nanometer to millimeters; 2) The identification of the relationship between the characteristic stiffness and packing density for scales beyond a few hundreds of nanometers (the observation scale); these relationships are insensitive to the varying mineralogy of the tested shale samples; (3) A clay packing density of $\eta \sim 0.5$ as a percolation threshold of the elasticity of the clay composite (the matrix in the shales is composed of the clay composite); and (4) A direct correlation between the clay packing density and the degree of anisotropy.

Bennett et al. (2015) coupled nanoindentation experiments with high-resolution imaging techniques, and simulated the heterogeneous scaled nanoindentation experiments using Nonlinear Finite Element Modeling (FEM). The indentation depth in these experiments were from

**Table 1a**
Experimental properties of nanoindentation-based testing on organic-rich shales.

| Experimental Properties | Ulm and Abousleiman (2006) | Bennett et al. (2015) | Wilkinson et al. (2015) | Abousleiman et al. (2016) | Liu et al. (2016) |
|---|---|---|---|---|---|
| Shale & locations | n/a | Woodford, USA | Bakken, USA | Woodford, USA | Bakken, USA |
| Coring depths (m) | n/a | 33–60 | Sample 1: 2195, Sample 2: 2134 | n/a | n/a |
| Coring Orientation to bedding | n/a | Parallel and Perpendicular | n/a | n/a | Parallel |
| Shape and no. Of samples | Cylindrical, 3 | n/a | 2 | n/a | Irregular, 4 |
| Avg. Sample dimensions (cm) | n/a | n/a | n/a | 1 | n/a |
| Sample thickness (mm) | 5–10 | n/a | n/a | 4 | n/a |
| Sample surface roughness ($\mu$m) | 0.09 | n/a | Sample 1: 0.01, Sample 2: 0.002 | n/a | n/a |
| Indenter tip diameter (nm) | n/a | n/a | Tip 1: 1180, Tip 2: 170 | 5000 and 10,000 | n/a |
| Depth of penetration ($\mu$m) | 0.2–0.3 | 0.2, 1, 3, and 5 | n/a | n/a | n/a |
| Maximum peak load ($\mu$N) | 236–266 | n/a | – | n/a | n/a |
| Quasistatic force[1] ($\mu$N) | – | – | 2 for both samples | – | – |
| Dynamic force[1] ($\mu$N) | – | – | Sample 1: 1, Sample 2: 0.5 With frequency of 200 Hz | – | – |
| Hold time for peak load (sec) | n/a | 60 | n/a | n/a | n/a |
| Time to load (sec) | n/a | n/a | n/a | n/a | n/a |
| Assumed sample Poisson's ratio | n/a | n/a | n/a | n/a | n/a |
| Loading rate: (nm/sec) for displacement controlled mode, ($\mu$N/sec) for force controlled mode | n/a | n/a | n/a | 15 (nm/sec) | 20 |
| Grit sizes used | n/a | 400, 600, 1500 | n/a | Up to 4000 and 1 $\mu$m diamond grit | 600 to 1200 and 5, 3, and 1 $\mu$m diamond grit |
| Porosity (%) | Sample 1: 37.3, Sample 2: 23.7, Sample 3: 13.9 | n/a | n/a | n/a (16–19) | n/a |
| Avg. TOC (wt.%) | n/a | n/a | Sample 1: 16, Sample 1: 14 | n/a | n/a |
| Dominant minerals | Clay minerals | n/a | n/a | Quartz and Clay minerals (Illite) | Quartz, Feldspar and Clay minerals |
| Temperature conditions (°C) | n/a | n/a | n/a | n/a | Ambient |
| Impact on geomechanical properties | Anisotropy, Clay packing density | Anisotropy | Polishing | Impact of OM on tensile strength | Mineral content, and Porosity |
| Remarks | First application of gridded nanoindentation | Comparison with FEM simulations, study of mechanical properties of both single mineral phase and composite phase | First application of modulus mapping | Experiment setup was positioned inside SEM | Measuring fracture toughness from nanoindentation |
| Limitations | | Not consider the geochemical characterization of OM and characterizing OM types (kerogen, bitumen, etc.) | Neglecting the impact of anisotropy | Not consider the possible impact of different OM types on tensile strength | Neglecting the impact of anisotropy |

[1] Quasistatic force and dynamic force are related to modulus mapping (Nano-DMA).

200 nm to 5 μm. The shallow indentation measured the homogeneous regions that contained a single mineral phase, while the larger indentations measured a composite deformation behavior. The specimens were made from the same sample, but the cutting direction was in two formats: bedding plane normal (BPN) and bedding plane parallel (BPP). The existing anisotropy of material properties was measured by conducting nanoindentations in directions both normal to the bedding plane (BPN) and parallel to it (BPP). Bennet et al. (2015) observed a significant creep in addition to plastic deformation in their nanoindentation experiments, so they applied a viscoplastic creep material for their FEM simulations. This constitutive model was specifically isotropic linear elasticity and a modified Drucker-Prager/Cap plasticity, and it was combined with the consolidation hardening and creep. This constitutive model provides an inelastic hardening mechanism to model the plastic compaction such as the consolidation of geological materials (Resende and Martin, 1985). For more information about this constitutive model readers are referred to Resende and Martin (1985), Liu and Borja (2013). The actual data from the nanoindentation experiments were used to calibrate the Drucker-Prager/Cap plasticity and consolidation creep material parameters. To understand the difference between the constitutive model parameters derived from BPN and BPP experiments, a triaxial compression test was simulated using those calibrated parameters.

From the above investigative approaches, Bennett et al. (2015) found three main mechanical phases, namely quartz-feldspar-pyrite (QFP), clays, and OM, from the shallow depth (200 nm) nanoindentation experiments. The FEM simulations of the triaxial compression test revealed anisotropic behavior of shale, that corresponds to the difference between the BPP and BPN calibrated constitutive model properties in the ductile stress-strain response. According to the authors, FEM simulations based on the completely isotropic material model showed that development of a finite strain anisotropic material model is necessary for shale at the nano-to micro-scales.

Abousleiman et al. (2016) focused on the failure analysis of OM-rich shales, and the possible impact of OM on the tensile and compressional strength at the nano-to micro-scales. They milled four micro-beams and three micro-pillars for their indentation experiments, and these samples were manufactured using the S.G. Roberts method (Di Maio and Roberts, 2005; Frazer et al., 2015). All the nanoindentation experiments were conducted inside a high-resolution SEM instrument, to provide a

**Table 1b**
Experimental properties of nanoindentation-based testing on organic-rich shales.

| Experimental Properties | Zargari et al. (2016) | Liu et al. (2018a) | Liu et al. (2018b) | Slim et al. (2019) | Zhao et al. (2019) | Shi et al. (2020) |
|---|---|---|---|---|---|---|
| Shale & locations | Bakken, USA | Bakken, USA | Bakken, USA | Marcellus, Haynesville, Antrim, Barnett and Woodford, in USA | Yanchang, China | Longmaxi, China |
| Coring depths (m) | n/a | n/a | n/a | Woodford sample is an outcrop | n/a | n/a |
| Coring Orientation to bedding | n/a | Perpendicular | Perpendicular | Parallel and Perpendicular | Parallel and Perpendicular | n/a |
| Shape and no. Of samples | Cubic, 4 | Irregular, n/a | Irregular, 4 | n/a | Cube, 7 | 2 |
| Avg. Sample dimensions (cm) | 1 | n/a | n/a | n/a | 1 | 0.5–1 |
| Sample thickness (mm) | 10 | n/a | n/a | 5–15 | 5 | n/a |
| Sample surface roughness ($\mu$m) | n/a | <0.2 | n/a | 0.030–0.150 | <0.1 | <0.07 |
| Indenter tip diameter (nm) | 160 | n/a | n/a | n/a | n/a | n/a |
| Depth of penetration ($\mu$m) | n/a | n/a | n/a | n/a | Microindentation: 4 | n/a |
| Maximum peak load ($\mu$N) | – | – | n/a | 4800 | Nanoindentation 3000 for minerals, 1000 for OM | 350,000 |
| Quasistatic force[1] ($\mu$N) | 2 | n/a | – | – | 2 | – |
| Dynamic force[1] ($\mu$N) | 1 with frequency of 200 Hz | n/a | – | – | 1 | – |
| Hold time for peak load (sec) | n/a | 60 | 5 to 180 | 180 | 2 | 0, 15, 60, 120, and 220 |
| Time to load (sec) | n/a | n/a | n/a | 10 | 10 | n/a |
| Assumed sample Poisson's ratio | n/a | n/a | n/a | n/a | n/a | 0.25 |
| Loading rate: (nm/sec) for displacement controlled mode, ($\mu$N/sec) for force controlled mode | n/a | n/a | n/a | 480 ($\mu$N/sec) Force-controlled nanoindentation | 300 ($\mu$N/sec) for minerals, 100 ($\mu$N/sec) for OM Force-controlled nanoindentation | n/a |
| Grit sizes used | n/a | n/a | 80 to 1200 | 400, oil-based diamond suspension, aluminum oxide abrasive disks (grain size between 9 and 1 $\mu$m) | n/a | 600 to 1200, and diamond polisher with grain sizes of 5, 3 and 1 $\mu$m |
| Porosity (%) | 3.7 to 6.9 | n/a | n/a | Marcellus: 5.9–8.4, Haynesville: 6–7.6, Antrim: 8.8, Barnett: 7.3, Woodford: 12.6 | n/a | n/a |
| Avg. TOC (wt.%) | 7.9 to 21.7 | n/a | n/a | Marcellus: 0.5–8.18, Haynesville: 2.6–3.3, Antrim: 9.6, Barnett: 12.2, Woodford: 4.2 | 8.68 to 13.27 | 2.37–6.51 |
| Dominant minerals | n/a | n/a | Quartz, feldspar, and clay minerals | Quartz and clay minerals | Quartz, feldspar, and clay minerals | Quartz and clay minerals |
| Temperature conditions (°C) | Ambient | n/a | n/a | Ambient | n/a | Ambient |
| Impact on geomechanical properties | Maturity, Porosity | Mineral content | Creep time, mineral content | Anisotropy, mineral content, OM | Anisotropy, mineral content, | Creep time, Creep rheological models |
| Remarks | | Nano-DMA can investigate creep | | Wide variety of samples | Comparison between gridded nanoindentation and modulus mapping | Compare three different rheological models |
| Limitations | Neglecting the impact of anisotropy | Not consider the geochemical characterization of OM and characterizing OM types, Neglecting the impact of anisotropy | Not consider the geochemical characterization of OM and characterizing OM types, Neglecting the impact of anisotropy | Characterizing OM types | Not consider the geochemical characterization of OM and characterizing OM types | Not consider the geochemical characterization of OM and characterizing OM types, Neglecting the impact of anisotropy |

[1] Quasistatic force and dynamic force are related to modulus mapping (Nano-DMA).

simultaneous monitoring of the indentation experiments until failure. The indenter tip was placed at the end of the free side of the micro-beams (Fig. 2), and the indentation experiments on the micro pillars were conducted to model the conventional unconfined compressional test at the nanoscale (Fig. 2). Moreover, a numerical simulation of the same experimental configurations was constructed to theoretically investigate the observed failure behaviors. The indentation experiments in Abousleiman et al. (2016) showed that the OM "glues together" other components of the shale, and it has a high modulus of toughness. In the work of Abousleiman et al. (2016), the OM-rich shales were directly loaded under tensile forces, but the classical macroscale experiments of tensile behavior did not measure or isolate the tensile strength directly.

Liu et al. (2018b) continued research by Liu et al. (2018a) and focused on the application of gridded nanoindentation (Constantinides et al., 2006) to understand the creep behavior of OM-rich shales. The holding phase of the nanoindentation was set to be 20s, 50s, 100s, and 180s. To find the best creep curve that fits the recorded data (indentation depth vs. creep time), Burger's model was applied. This is one of several rheological models for investigating the creep behavior (Ashraf and Tian, 2016; Cornet, 2015). The impact of various existing minerals on the creep behavior and mechanical properties (Young's modulus, hardness, and fracture toughness) was studied using bivariate plots from partial least-squares regression (PLS) (Table 6 in Liu et al., 2018b). Based on this analysis, it was found that a change in the clay content has the greatest effect on the mechanical properties. The increase in clay mineral content leads to an increase in changes to the Young's modulus and hardness, but the change of fracture toughness is reduced. They applied the energy analysis method (Cheng et al., 2002; Liu et al., 2016; described in Section 2.4 of this work) to calculate the fracture toughness value for each sample. It was shown that as the creep time increases, the fracture toughness decreases.

Slim et al. (2019) investigated the potential effect of OM on the creep behavior of a source rock using both experimental and modeling methods. A wide variety of shales in the US, including Marcellus, Haynesville, Antrim, Barnett, and Woodford, was investigated. Based on the data from Rock-Eval pyrolysis analysis, Haynesville shale samples are mature (vitrinite reflectance equivalent of 2.5%), but Woodford, Barnett, and Antrim shale samples are immature source rocks. Moreover, the Marcellus shale samples are overmature source rocks (vitrinite reflectance equivalent of 3%). A mechanical test and chemical analysis were also conducted on pyrobitumen-rich samples to understand the creep properties of mature OM. A grid indentation technique (Constantinides et al., 2006; Ulm et al., 2007) was applied, given the heterogeneity of the shale samples. Slim et al. (2019) identified four mechanical phases within the shale sample and reported that these mechanical phases are based on the similarity in mechanical strength or stiffness properties. The shale components whose mechanical strength or stiffness properties are in the same range, are categorized in the same mechanical phase. The four mechanical phases are: 1) OM-rich phase, 2) a porous OM phase with inorganic impurities, 3) a mixed phase (containing pyrite), and 4) an inclusion (quartz) phase (Figs. 5 and 6 in Slim et al., 2019). Of these four identified phases, the highly porous organic-rich phase has the lowest values of *M*, *H*, and *C*, and an inclusion phase has the highest values of M, H, and C (Fig. 5 in Slim et al., 2019). M and H are indentation modulus and indentation hardness, respectively (explained in Section 2.1.1). C is the contact creep modulus (for more detailed information about C, readers are referred to Vandamme and Ulm, 2013). The OM makes the greatest contribution to the creep behavior of the shale, but the inorganic phases surrounding OM reduce the creep rate. A comparison of the results from microindentation tests with first-order creep homogenization modeling shows that two asymptotes exist for the relevant creep rate of OM-rich shales: (1) a continuous matrix of immature OM with rigid clay inclusions defines the lower end-member; (2) a self-consistent pattern representing mature disordered systems defines an upper end-member.

Zhao et al. (2019) investigated compositional heterogeneity and anisotropy of shales using the mechanical characterization method (nanoindentation and modulus mapping) at three (mineral, core, and

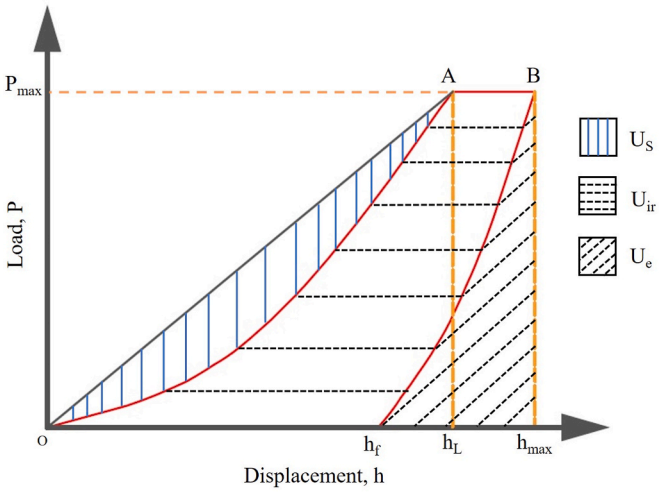



**Fig. 3.** Schematic of the energy analysis in the nanoindentation experiments. $U_S$ is the absolute work which can be calculated from the area of $OAh_LO$, and $U_e$ is the elastic work defined as the area under the curve of the unloading stage ($h_fBh_{max}h_f$). Furthermore, $U_{ir}$ is the irreversible or the plastic energy work determined by the area of $OABh_{max}O$ (Jha et al., 2012).

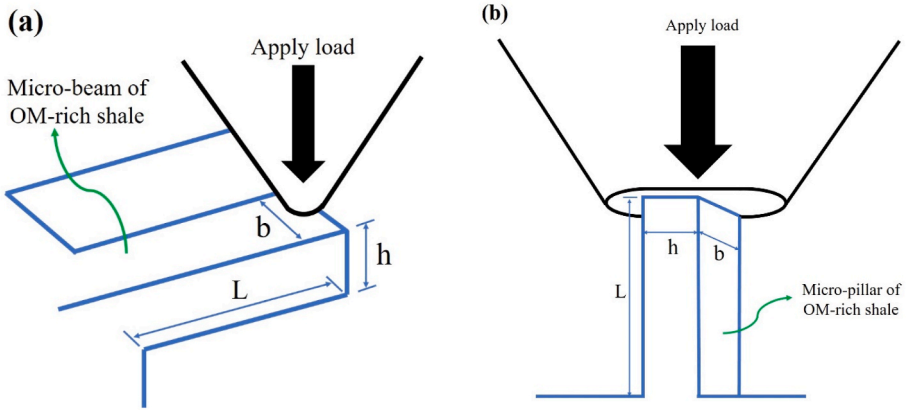



**Fig. 2.** Schematic of the nanoindenter, (a) micro-beam and (b) micro-pillar of OM-rich shales. (a) Nanoindenter is positioned at the end of the micro-beam. (b) Nanoindenter is positioned on top of the micro-pillar.

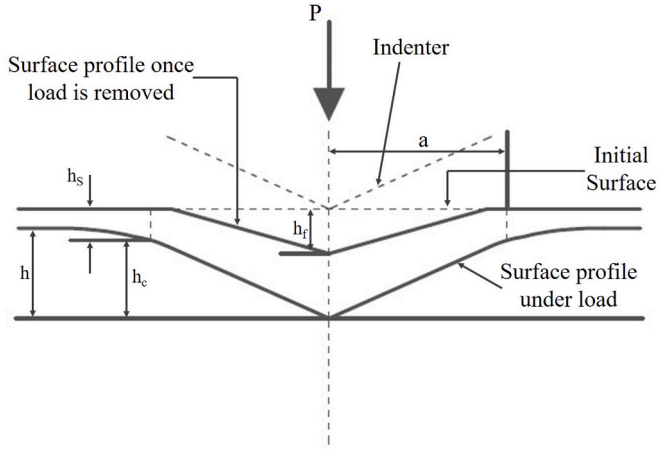



**Fig. 4.** Schematic of the unloading stage in a nanoindentation experiments (after Oliver and Pharr, 1992).

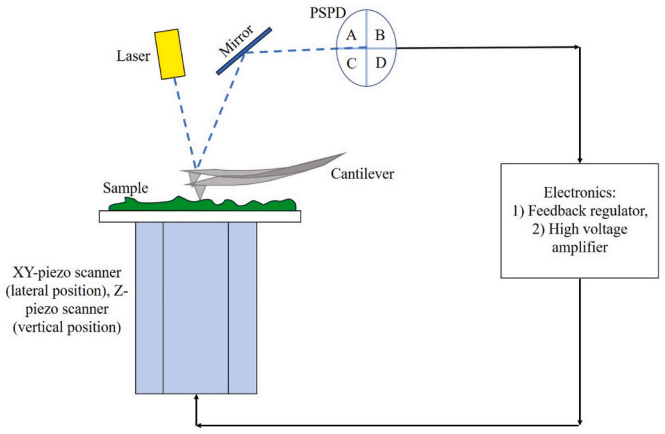



**Fig. 5.** General schematic of a typical AFM. A reflected laser from the cantilever is detected by the quadrant photodiode (PSPD), and a signal is transmitted to the data acquisition equipment.

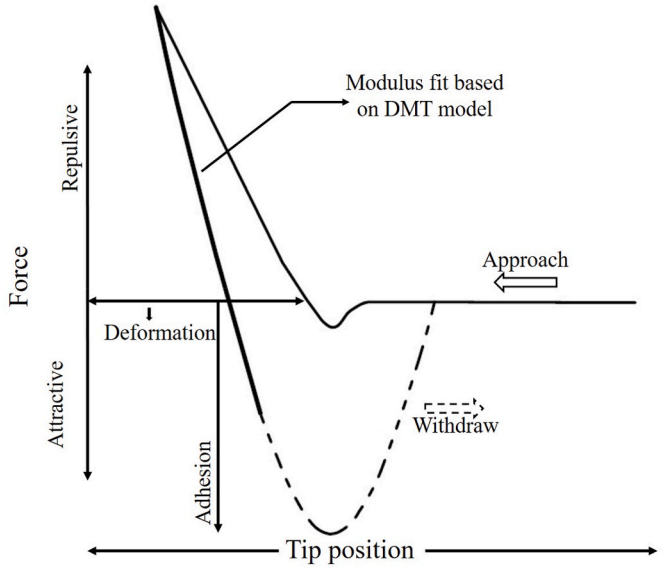



**Fig. 6.** Schematic of a generated force-distance curve for a single tapping of AFM.

well) scales. Seven terrestrial shale samples from the Yanchang formation in China were chosen. Nanoindentation and modulus mapping were applied to investigate the compositional heterogeneity and anisotropy, respectively, at the mineral scale. The Young's modulus and hardness of non-clay minerals were determined by nanoindentation; to study anisotropy, the mechanical properties of clay in two directions were measured with modulus mapping. Microindentation was performed to verify the prediction of the upscaling method, which was proposed to predict the macroscopic characteristics of organic-rich shale from the mineral scale properties. Both microindentation and nanoindentation were based on the same theories, and the difference was just in the scale of these experiments. The upscaling method was based on both the Mori-Tanaka and Self-Consistent schemes. The well scale mechanical characterization was based on the Brittleness Index (BI), newly proposed by Zhao et al. (2019) which is a function of the mineral composition, and its application to interpretation of well logs.

Shi et al. (2020) conducted gridded nanoindentations on 6 × 6 grids of the nanoindentation map with 15 μm as the distance between two neighboring nanoindentations. They chose two shale samples from the upper Ordovician to Silurian Longmaxi black shale, Sichuan Basin, China for gridded nanoindentation experiments. The holding phase of the nanoindentation was set to 0s, 15s, 60s, 120s, and 220s. Shi et al. (2020) used three different viscoelastic rheological models, namely the i) three-element Voigt, ii) Burgers, and iii) two-dashpot Kelvin models. These three models were fitted with the depth-time curve during the nanoindentation holding phase. For detailed explanation and formulation of these three rheological models, readers are referred to Cornet (2015), Ashraf and Tian (2016), and Shi et al. (2020).

### 2.3. Modulus mapping application

Wilkinson et al. (2015) introduced an elastic modulus mapping technique, using spatially continuous dynamic nanoindentation. Two shale samples from the Bakken Formation in the Williston Basin, North Dakota were used. The samples were sectioned and polished using a focused gallium ion beam. Two samples were polished differently to investigate the impact of sample preparation on the recorded stiffness data and mechanical characterization of samples by dynamic nanoindentation.

The modulus map, surface topography and displacement amplitude of each sample were presented in Wilkinson et al. (2015). There are two important issues with the modulus maps: (1) The ion tracks are presented as the change of modulus within a relatively mineralogically homogeneous region. Moreover, the surface height is changed drastically because the softer material can be removed to a greater extent than the harder minerals; these local changes in topography can lead to an under- or over-estimation of contact area between the sample surface and the tip. (2) The displacement amplitude (refer to Eq. (3) in this work) in the stiff minerals (mainly pyrite) can be lower than the detection limit of the instrument. Therefore, the system counts them as noise which can result in erroneous values for the modulus. Comparison of the results of both shale samples shows that the sample preparation (polishing method) has a direct impact on the recorded modulus maps. The importance of this research lies in its investigation of the possible effect of large surface height variations, and low displacement amplitudes on the results of modulus mapping.

Using a coupled method of nano-DMA and gridded nanoindentation, Liu et al. (2018a) concentrated on the time-dependent mechanical behavior of shales. Based on the recorded complex modulus and hardness values, they selected three points on the gridded nanoindentation area to represent soft minerals, minerals with medium strength, and hard minerals, respectively. Based on the plotted creep displacement vs. time for these three points (Fig. 8 in Liu et al., 2018a), the displacement in all selected points abruptly increased at the beginning of the creep and then slowed down. Furthermore, this figure showed that the material with a greater hardness and more complex modulus has the lower

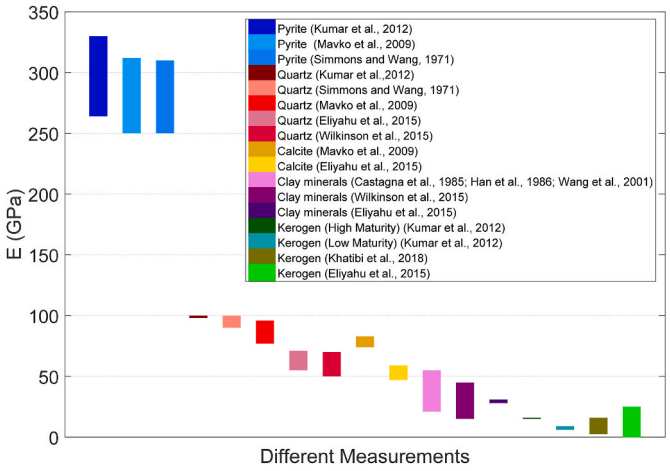



**Fig. 7.** Young's modulus of important constituent minerals in OM-rich shales.

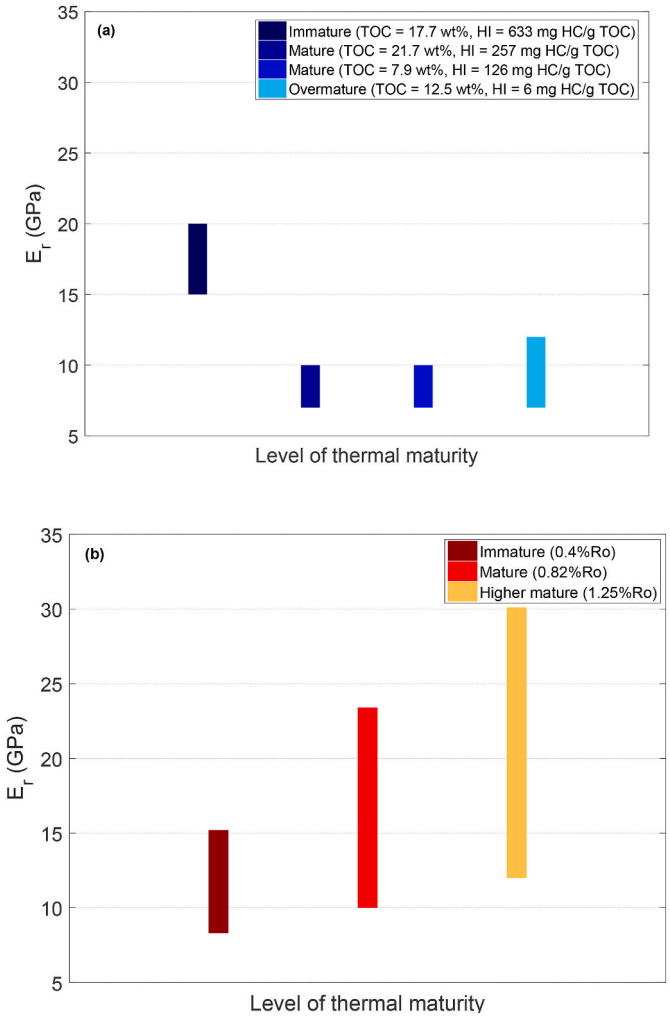



**Fig. 8.** Reduced elastic modulus of kerogen at different maturity levels: (a) the results from Zargari et al. (2013), and (b) the results from Emmanuel et al. (2016a).

creep displacement.

Liu et al. (2018a) also measured the average value of creep displacements, loss modulus, storage modulus and hardness in the entire nanoindentation area. These average values were measured for each time (during the creep experiment) to understand the overall creep behavior in the entire gridded areas. Average creep displacement has the same behavior through time at three selected points. The curve fitting analyses on creep displacement, loss modulus, storage modulus and hardness showed a logarithmic pattern. The groups of soft and hard minerals are distinguishable from the resultant histogram of storage modulus map from the deconvolution method. From Table 6 in Liu et al. (2018a), it is clear that the calculated volumetric fraction of soft minerals increased throughout the creep experiment, but the fraction of hard minerals decreased as times passed. These authors provided two possible scenarios for this behavior: i) the impact of dislocation creep, and ii) the movement of hard minerals under the soft minerals close to the sample surface.

### 2.4. The relationship of fracture toughness and elastic properties at the nanoscale

A nanoindentation experimental technique has been previously applied to measure the fracture toughness of many materials other than shale (Evans and Charles, 1976; Jaya et al., 2015; Lawn et al., 1980; Merle and Göken, 2011). Jha et al. (2012) described an energy-based analysis to measure the fracture toughness from their nanoindentation experiments:

Young's modulus, hardness, and contact stiffness are calculated from the force-displacement curve of the nanoindentation experiments (Fig. 1).

$$S = \left(\frac{dP}{dh}\right)_{h=h_{\max}} = 0.75 \times (2\nu_E - 1)\frac{P_{\max}}{h_{\max}} \tag{7}$$

where $\nu_E$ (elastic energy ratio) is defined as the ratio of the absolute work over the elastic work:

$$\nu_E = \frac{U_S}{U_e} \tag{8}$$

$U_e$ is the elastic work to be defined as the area under the curve of the unloading stage ($h_f B h_{max} h_f$) (Fig. 3), and $U_S$ is the absolute work which can be calculated from the area of $OAh_LO$. Moreover, the $U_{ir}$ is the irreversible or the plastic energy work determined by the area of $OABh_{max}O$, and the total work is defined by the summation of the plastic and elastic work.

Fig. 4 shows the unloading stage of the indentation tests, and $h_{max}$ is calculated as follows:

$$h_{\max} = h_s + h_c \tag{9}$$

where $h_c$ is the contact depth, and $h_s$ is the vertical displacement of the surface at the circumference of the contact. According to Oliver and Pharr (1992), $h_c$ and $h_s$ can be calculated by Eqs. (10) and (11) respectively:

$$h_c = \frac{2(\nu_E - 1)}{(2\nu_E - 1)}h_{\max} \tag{10}$$

$$h_s = \varepsilon\frac{P_{\max}}{S} \tag{11}$$

where $\varepsilon$ is the geometric constant, with values of 0.72, 0.75, and 1 for the conical indenter, paraboloid of revolution and the flat punch, respectively.

After deriving the contact depth (Eq. (10)) and initial contact stiffness (Eq. (7)), the reduced Young's modulus can be derived based on Oliver and Pharr (1992):

$$E_r = \frac{\sqrt{\pi}S}{2\sqrt{A_C}} \tag{12}$$

where $A_C$ is the area of contact which is calculated from

$$A_C = 24.5h_c^2 \tag{13}$$

Finally, the hardness and Young's modulus can be derived from Eqs. (1) and (6), respectively. From these calculations based on nanoindentation, it is possible to calculate the fracture toughness. The fracture energy ($U_{Crack}$) can be obtained by

$$U_{Crack} = U_{ir} - U_{pp} \tag{14}$$

where $U_{ir}$ is the irreversible energy, which is calculated by the difference between the total energy ($U_t$) and the elastic energy ($U_e$) based on Cheng et al. (2002). $U_{pp}$ is the energy of pure plasticity, and it is derived from Eq. (15) of Cheng et al. (2002):

$$\frac{U_{pp}}{U_t} = 1 - \left[\frac{1 - 3\left(\frac{h_f}{h_{\max}}\right)^2 + 2\left(\frac{h_f}{h_{\max}}\right)^3}{1 - \left(\frac{h_f}{h_{\max}}\right)^2}\right] \tag{15}$$

Consequently, the critical energy release rate ($G_C$) can be derived using the following equation:

$$G_C = \frac{\partial U_{Crack}}{\partial A} = \frac{U_{Crack}}{A_{\max}} \tag{16}$$

where $A_{max}$ is the maximum area of the crack, which is calculated as $A_{max} = 24.5h_{max}^2$, for the Berkovich indenter. The critical energy release rate ($G_C$) and the reduced modulus ($E_r$) are used to calculate the fracture stress intensity factor ($K_C$) as follows:

$$K_C = \sqrt{G_C E_r} \tag{17}$$

Liu et al. (2016) applied nanoindentation to measure the fracture toughness of OM-rich shales. They applied an energy analysis method because of the inherent inhomogeneity of the shale samples, which made the measurement of the crack, or fracture, length difficult during the nanoindentation process. The best fit curve to the loading and unloading stages had a power law function (for the curve fitting parameters, readers are referred to Table 1 in Liu et al., 2016). They used an extrapolation method of Fischer-Cripps (2006) to determine the Young's modulus and hardness for each test. The relationships between the clay packing density and Young's modulus and hardness were also investigated. Liu et al. (2016) concluded that porosity, as compared to clay content, has a greater influence on the mechanical properties, and this agrees with the multivariate linear regression analysis of Kumar et al. (2012). The results of 20 nanoindentation tests were used to derive the critical energy release rate and the fracture stress intensity factor. Liu et al. (2016) found a linear relationship between the fracture toughness and Young's moduli (Eq. (17) in Liu et al., 2016). As there are few published papers on the fracture toughness of shales, this relationship can provide a method for its determination.

### 2.5. Summary of the application of nanoindentation-based methods to OM-rich shales

Ulm and Abousleiman (2006) found the linear relationship between the measured indentation modulus (for both nanoindentation normal to the bedding planes and parallel to the bedding planes) and clay packing density for clay packing densities greater than 0.5. The difference between measured indentation modulus for nanoindentations normal to the bedding planes and nanoindentations parallel to the bedding planes becomes larger when the clay packing density approaches 1. These findings shows that the main source of existing anisotropy of shale elasticity parameters is related to the depositional structure of the clay grains. The capability of nanoindentation to measure the fracture toughness in OM-rich shales was studied by Liu et al. (2016). Their main focus was to find the relationship between the measured Young's modulus and fracture toughness. It is possible to investigate the capability of nanoindentation on measuring the fracture toughness of OM, which is the most compliant compound of OM-rich shales comparing to other constituent minerals of shales.

The impact of bedding on the stiffness measurements and its anisotropy was studied by Bennett et al. (2015), Li et al. (2019), Slim et al. (2019). Table 2 shows the results of measured mechanical properties of nanoindentations at two different depths (200 nm and 1 μm) from Bennett et al. (2015). Both the effective modulus ($E_S$) and indentation hardness ($H$) of the QFP are higher when measured parallel with the bedding planes than the normal to the bedding planes. For clay and O/C (organic and clay) matrices, the effective modulus parallel with the bedding plane is higher than that normal to the bedding planes, but the indentation hardness normal to the bedding is higher than that parallel with the bedding plane. It is important to mention that Bennet et al. (2015) observed four constituent compounds in their shale samples including clay particles, QFP, OM, and O/C (composite of clay and organic matter). In the O/C compound the clay particles and organic matter were mixed without forming a supporting skeleton with inter-granular contacts. The OM has the highest anisotropy ratio, while clays and QFP are second and third, respectively. Slim et al. (2019) found that the creep rate of source rocks is mostly isotropic while their elastic properties show anisotropic behavior. According to Slim et al. (2019), isotropic constituents, pores, and OM are the deriving parameters in the isotropic creep behavior of organic-rich shales.

The impact of OM maturity on the stiffness measurements using modulus mapping was studied by Zargari et al. (2016). They found that as OM maturity increased, the lower the moduli of the kerogen were. It was mentioned that OM with higher maturity has an abundant nanoporosity and concluded that the intraparticle OM porosity has a crucial impact on its modulus.

Shi et al. (2020) investigated the creep behavior of OM-rich shales using gridded nanoindentation coupled with three rheological models of creep. A creep time constant was derived from each of the rheological models, and the regression precision of these three models for shale creep was compared. Even though the two-dashpot Kelvin model had the highest standard deviation, it provided better regression results than the three-element Voigt and Burgers models. As Slim et al. (2019) found that the OM has the greatest contribution to the creep behavior of the shale, it is important to investigate the creep behavior of OM more in detail such as investigating whether or not different types of OM have different creep behavior or not, or investigate the relationship between the maturity and creep behavior of OM. Reviewing most of the published research about application of nanoindentation in the mechanical characterization of OM-rich shales showed that most of them are lacking in identifying the types of OM (kerogen, bitumen, etc.), and finding the possible relationship between the OM type and mechanical properties such as Young's modulus and Creep rate is difficult. Due to the small scale of the indentation depth, gridded nanoindentation is not capable of identifying the possible impact of some internal structures of OM-rich shales, such as natural fractures and large pores. Therefore, to understand the possible impact of natural fractures and large pores, it is necessary to combine gridded nanoindentation with a good upscaling method to find the homogenized values of mechanical properties (Young's modulus, hardness, etc.). Moreover, it is necessary to compare the upscaled mechanical properties with the results of conventional rock mechanics tests on the same type of samples.

## 3. Atomic force microscopy (AFM)

Introduced for the first time by Binnig et al. (1982), the Scanning-tunneling microscopy (STM) was the precursor to the current version of AFM. Binnig et al. (1986) extended STM applications to scan surfaces other than conductive and semiconductive materials, and this resulted in the invention of AFM (see Fig. 5 for a general schematic). The tip was mounted on a cantilever spring, and the monitoring system controlled the cantilever deflection corresponding to the tip/surface

**Table 2**
Results of nanoindentations in two different depth (200 nm and 1 μm) from Bennett et al. (2015). The specimen was made from the same sample, but the cutting direction was in two directions: bedding plane normal (BPN) and bedding plane parallel (BPP). $K_{(.)}$ is the anisotropy ratio of the mechanical property (.), which can be $E_S$ (effective modulus), $H$ (hardness).

| | BPN | BPP | $K_{(.)}\left(K_{(.)}=\frac{(.)_{BPP}}{(.)_{BPN}}\right)$ | | BPN | BPP | $K_{(.)}\left(K_{(.)}=\frac{(.)_{BPP}}{(.)_{BPN}}\right)$ |
|---|---|---|---|---|---|---|---|
| Indentation depth = 200 nm | | | | | | | |
| QFP | | | | Clay | | | |
| $E_S(GPa)$ | 32.750 | 38.114 | 1.16 | $E_S(GPa)$ | 20.373 | 25.865 | 1.27 |
| $H(GPa)$ | 4.515 | 4.815 | 1.07 | $H(GPa)$ | 1.801 | 1.592 | 0.88 |
| Indentation depth = 200 nm | | | | Indentation depth = $1\mu m$ | | | |
| Organic matter | | | | O/C matrix | | | |
| $E_S(GPa)$ | 8.780 | 6.503 | 0.74 | $E_S(GPa)$ | 12.392 | 18.545 | 1.50 |
| $H(GPa)$ | 0.449 | 0.278 | 0.62 | $H(GPa)$ | 0.604 | 0.592 | 0.98 |

forces (Binnig et al., 1986). As the tip/surface interaction forces are a function of gap width, the AFM feedback loop can keep the gap between tip and surface constant using vertical tip displacement compensating for cantilever deflection.

### 3.1. Nanoscale mechanical mapping using AFM

Nanoscale mechanical mapping using AFM can be conducted in two different modes: the first mode applies a relatively large loading force, so it triggers both elastic and plastic deformations. The second mode applies a relatively small loading force, so it can just measure the elastic deformation. The AFM manufacturers assign a different name to the force mapping modes. For instance, Bruker name it PeakForce quantitative nano-mechanical mapping (PeakForce QNM™), while Park Systems call it PinPoint™. Nanoscale mechanical mapping has been conducted by AFM using a PeakForce QNM™ method (Pittenger et al., 2014; Su, 2010). It is a relatively new AFM mode derived from the PeakForce Tapping™ method, in which the Z piezo oscillates at a rate far lower than the probe resonance (Pittenger et al., 2014; Su, 2010). The sample surface topography and force-distance curve are identified for each image pixel. The normal deflection of the cantilever is used for the surface topography measurement. To maintain a constant imaging force, the controlling signal is the maximum force of each tap. Fig. 6 shows the schematic diagram of a generated force-distance curve for a single tap of the PeakForce QNM™ technique. In this method, the reduced elastic modulus ($E^*$) can be calculated from the Derjaguin-Muller-Toporov (DMT) model (Derjaguin et al., 1975; Trtik et al., 2012):

**Table 3**
Experimental properties of AFM-based testing on organic-rich shales.

| Experimental Properties | Eliyahu et al. (2015) | Emmanuel et al. (2016a) | Emmanuel et al. (2016b) | Li et al. (2018b) |
|---|---|---|---|---|
| Shale & locations | n/a | n/a | n/a | Bakken, USA |
| Coring depths (m) | n/a | n/a | n/a | 2500–3300 |
| Coring Orientation to bedding | n/a | n/a | n/a | Parallel |
| Shape and no. Of samples | n/a, 3 | n/a, 6 | n/a, 2 | n/a, 4 |
| Avg. Sample dimensions (cm) | n/a | n/a | n/a | n/a |
| Sample thickness (mm) | n/a | n/a | n/a | n/a |
| Sample surface roughness ($\mu$m) | n/a | n/a | n/a | 0.340 |
| AFM probe tip diameter (nm) | 40 | 8–12 | 8–12 | n/a |
| Probe normal spring constant ($N\,m^{-1}$) | 650 | 200 | 200 | 200 |
| Assumed sample Poisson's ratio | 0.3 | 0.4 | 0.4 | 0.2–0.5 |
| Grit sizes used | n/a | n/a | n/a | 1200 and 3000 |
| Porosity (%) | 4.7–10.9 | n/a | n/a | n/a |
| Avg. TOC (wt.%) | 0.52–4.56 | 0.76–11.21 | 3.23–6.42 | 9.04–16.3 |
| Dominant minerals | Quartz, Clay Minerals | Quartz, carbonate Minerals | Quartz, carbonate Minerals | Quartz, clay Minerals |
| Temperature conditions (°C) | Ambient | Ambient | 25–225 | n/a |
| Impact on geomechanical properties | Poisson's ratio | Thermal maturity | Temperature | Poisson's ratio, Thermal maturity |
| Remarks | First application of PeakForce QNM™ on OM-rich shales | | Impact of reservoir temperature on stiffness of OM (both kerogen and bitumen) | |
| Limitations | Neglecting the impact of anisotropy, Characterizing OM types (kerogen, bitumen, etc.) | Neglecting the impact of anisotropy | Neglecting the impact of anisotropy | Neglecting the impact of anisotropy, Characterizing OM types (kerogen, bitumen, etc.) |

$$F_{interaction} = \frac{4}{3}E^{*}\sqrt{R(d-d_0)^3} + F_{adh} \tag{18}$$

where $F_{interaction}$ is the interaction force between the tip and sample, $d_0$ is the surface rest position, $(d - d_0)$ is the deformation of the sample, $R$ is the radius of the tip, and $F_{adh}$ is the adhesion force throughout the contact. There is a relationship between the Young's modulus of the sample $(E_s)$ and the reduced modulus of the sample $(E^*)$ thus:

$$E^{*} = \left(\frac{1-\nu_s^2}{E_s} + \frac{1-\nu_{tip}^2}{E_{tip}}\right)^{-1} \tag{19}$$

where $E_{tip}$ is the Young's modulus of the tip material, while $\nu_s$ and $\nu_{tip}$ are Poisson's ratio of the sample and probe, respectively.

### 3.2. AFM applications for nanomechanical characterization of shales

Table 3 represents the experimental properties of AFM applications by different researchers to investigate the mechanical behavior of OM-rich shales. Limitations in their experimental techniques are presented. Eliyahu et al. (2015) used the new AFM method (PeakForce QNM™, as described above) to map the modulus values of three samples from an Upper Jurassic source rock in the USA. The total organic carbon (TOC) contents of their samples ranged from 0.5 to 4.5 wt%, and their thermal maturity was very high (vitrinite reflectance ~2.1 %$R_o$). The scan area was changed from 5 $\mu m$ × 5 $\mu m$ to 50 $\mu m$ × 50 $\mu m$, and the values of modulus exhibited the same pattern and distribution in all of these different scan sizes, and clays and quartz grains had modulus values of approximately 30 GPa and 60 GPa, respectively. Furthermore, the map at highest resolution showed that the modulus values of the OM was in the range 0–25 GPa; more importantly, there were two distinct peaks in the distribution of the modulus values for the OM. This led to the conclusion that the OM probably constitutes two distinct types.

Emmanuel et al. (2016a) investigated the impact of thermal maturation on mechanical properties (stiffness) of OM at the nanoscale, using six OM-rich shale samples with three different levels of thermal maturity ($\%R_o = 0.40, 0.82, 1.25$) (Table 1 in Emmanuel et al., 2016a), which corresponds to immature to mature stages from oil to gas window generation. The samples were from a Cretaceous carbonate-rich source rock from the southern USA. The bulk mineralogy and OM properties were evaluated by standard organic petrology point count methods, Rock-Eval pyrolysis, and XRD. After polishing the samples, an Environmental SEM (ESEM) was used to identify areas that contained an OM particle. Based on the morphological criteria for OM that were introduced by Loucks and Reed (2014), bitumen is the OM that existed inside the fossil body-cavities, or the OM that existed within the voids that had been partially surrounded by mineral cementation. The other type of OM was identified as kerogen. Emmanuel et al. (2016a) used Loucks and Reed (2014) morphological criteria to identify kerogen and bitumen. AFM scans were conducted using the PeakForce QNM™ method to map the stiffness of the areas of interest (Eliyahu et al., 2015). Finally, the EDS (energy dispersive spectroscopy) detector was used to image, and map the chemical elements of, the sample surface. They assumed a Poisson's ratio value of 0.4 for organic matter. The Young's modulus of kerogen was found to be in the range of 3.2–29.9 GPa (Table 2 in Emmanuel et al., 2016a), which was in good agreement with the results in previously published papers such as Kumar et al. (2012) and Eliyahu et al. (2015). The measured reduced moduli of bitumen did not show the same trend as the reduced moduli of kerogen as function of maturity, and it was relatively uniform from low to high maturity. The average measured reduced moduli of bitumen at medium and high maturity levels are 8.7 GPa, and 9.8 GPa, respectively, in a good agreement with the published values of Young's modulus values for bitumen of 5–10 GPa (Eliyahu et al., 2015; Zargari et al., 2013).

Yang et al. (2017) introduced an application of AFM-based infrared spectroscopy (AFM-IR) to measure the mechanical and chemical heterogeneity of OM in shale at the nanoscale. AFM-IR is a novel technique in life and materials sciences, and it generates both chemical mapping and mechanical stiffness mapping at nanoscale, in addition to the surface topography image from AFM (Dazzi et al., 2010, Dazzi et al., 2012; Khanikaev et al., 2016; Qin et al., 2016). This was the first application of AFM-IR to the nanoscale characterization of shales. The samples are from the Middle Devonian-Lower Mississippian New Albany Shale, USA. A series of different samples were prepared with different maturities using laboratory hydrous pyrolysis, and these samples were subjected to the same AFM-IR analyses to understand the possible effect of maturation on mechanical and chemical properties of OM. Yang et al. (2017) demonstrated the change in individual organic macerals as a result of maturation, and they provided a microscopic image of the heterogenous process of petroleum generation.

AFM-IR works as follows: an AFM probe is placed in contact with an area of interest on a surface. The tip radius should be smaller than 25 nm. At the same time, the area is irradiated with a tunable IR laser (Dazzi et al., 2012). The sample thermally expands as a result of conversion of the energy of absorbed photons to heat. This expansion happens when the laser wavelength reaches the infrared absorption spectrum of the sample. This thermal expansion makes the AFM cantilever oscillate. The oscillation amplitude is a direct function of the local IR absorption coefficient at the IR wavelength where the sample starts to irradiate. As the cantilever and tip stay in contact with the sample surface, the contact resonance (CR) frequency of the sample and AFM cantilever is directly related to the sample's mechanical modulus (Dazzi et al., 2010, 2012; Dazzi and Prater, 2017). This relationship between the CR frequency and sample stiffness, which comes from the connection between the AFM cantilever oscillation amplitude and frequency, allows mapping of the IR absorption and stiffness simultaneously at a fixed laser wavelength.

Yang et al. (2017) analyzed a distribution of measured CR frequencies on different types of organic macerals (inertinite, solid bitumen and tasmanite). The maceral identification were conducted in epi-fluorescence and reflected white light under oil immersion, and reflectance measurements of solid bitumen. These macerals were also identified in stiffness images. Fig. 3 and Table 2 in Yang et al. (2017) provide information on these constituents' mechanical properties. According to this table, the average CR frequency of inertinite is 182.1 kHz ± 1.7 kHz, and the resulting average CR frequencies of solid bitumen and tasmanite are 178.8 kHz ± 3.6 kHz and 177.2 kHz ± 4.3 kHz, respectively. The spatial resolution of mechanical and chemical maps of the macerals from AFM-IR method are orders-of-magnitude smaller than the regular diffraction-limited IR microscopy (Yang et al., 2017). Combining the geochemical and geomechanical analyses of macerals also demonstrated that the mechanical properties of macerals, mainly their stiffness, are a function of their chemical composition. The maceral enriched in aromatic carbon (inertinite) is relatively stronger and has a relatively higher mechanical stiffness than bitumen or tasmanite.

Li et al. (2018b) used PeakForce QNM™ to measure the mechanical property (mainly stiffness) of four OM-rich shale samples from the Bakken formation. They found three distinct mechanical phases including a) a soft phase with a modulus of $< 25GPa$, b) a medium strength phase (modulus of $25 - 50GPa$), which is an intergranular matrix phase, c) a stiff phase, which is inorganic and mainly isolated grains with a stiffness of $50 - 100GPa$. Moreover, they found that the change in texture of OM can be the reason for difference between the Young's modulus of OM from one sample to the other.

Poisson's ratio is highly important to the measurement of Young's modulus in the PeakForce QNM™ method (Eq. (19) in this work). There are little data on the Poisson's ratio of OM (Ahmadov, 2011; Eliyahu et al., 2015); therefore, the possible effect of a change in Poisson's ratio (0.2–0.5) on the Young's modulus was studied in Li et al. (2018b). They found that a change in Poisson's ratio can lead to 4–25% difference between Young's modulus and reduced modulus values, and this is in agreement with the findings of Eliyahu et al. (2015), who noted an

uncertainty of up to 12% in the calculated Young's modulus of OM.

Tian et al. (2018) coupled X-ray photoelectron spectroscopy (XPS), Fourier transform infrared spectroscopy (FTIR) and AFM to understand the relationship between the chemical composition and adhesion force of OM. The measured adhesion forces of OM within the lacustrine and marine shale samples were $100.86 \pm 5\ \mu N$ and $94.28 \pm 4.7\ \mu N$, respectively. Their measurements showed that the OM in both continental shales and marine shales has a significantly higher adhesion force compared to inorganic components within the shale. They did not observe a significant change in the measured adhesion force when the preloading force was changed. Based on their results from XPS and FTIR, they suggested that the C═C and C═O bonds are the reasons for the large dipole on the OM surface. Tian et al. (2019) extended the work of Tian et al. (2018) to clay minerals in the shale, and compared the adhesion force and the existing chemical bonds in kerogen and clay minerals. The results from FTIR and XPS analysis of kerogen and clay minerals showed that the chemical bonds on the kerogen surface were O–H, C–C, C–O, pyrrolic, while the chemical bonds on the surface of illite were mainly Si–O and Al–O. The adhesion forces of other minerals such as calcite, montmorillonite and muscovite were $23.8 \pm 11.8$, $33.7 \pm 6.28$, and $105.1 \pm 9.1$ nN, respectively. Tian et al. (2020) treated the surface of shales with carbonic acid and investigated the impact of this treatment on the mechanical properties using SEM/EDS, AFM and nanoindentation. The measured adhesion force of the nanoindentation experiments revealed that the carbonic acid treatment made the samples more ductile, because the brittle carbonate minerals were highly prone to an acidic treatment.

Fig. 7 shows the experimentally measured Young's modulus for different components of shale from published research (Castagna et al., 1985; Eliyahu et al., 2015; Han et al., 1986; Khatibi et al., 2018; Kumar et al., 2012; Mavko et al., 2009; Simmons and Wang, 1971; Wang et al., 2001; Wilkinson et al., 2015). Most of these measurements were conducted at the nanoscale to microscale using nanoindentation or AFM. It is clear from Fig. 6 that OM is the least stiff component in shales ($< 30GPa$) and the pyrite is the most rigid constituent ($> 250\ GPa$).

### 3.3. Summary of application of AFM-based methods on OM-rich shales

AFM's capability to measure the mechanical properties of OM-rich shales has been the subject of several papers (Eliyahu et al., 2015; Emmanuel et al., 2016a, 2016b; Yang et al., 2017; Li et al., 2018b). It was shown that this mode of AFM can map the measured reduced modulus, then the map of measured reduced modulus was compared with the chemical spectral analysis of the scanned surfaces from SEM/EDS. Therefore, it is possible to use AFM-based nanomechanical characterization of shale to classify OM-rich shales based on stiffness measurements. Even though the impact of maturity on the stiffness and mechanical strength of OM was the subject of some AFM-based and nanoindentation-based published research, it is still unclear from this work whether the increase in maturity results in an increase or decrease in the stiffness of OM.

The impact of OM maturity on the stiffness measurements using the AFM-based method was studied by previous papers (Emmanuel et al., 2016a; Eichmann et al., 2018). Fig. 8 shows the measured reduced elastic modulus for OM at different maturity levels from Emmanuel et al. (2016a) and Zargari et al. (2016), which was based on modulus mapping. There is a big difference between the results of these two previous papers. The moduli of OM increased from the sample with the lowest maturity to the sample with the higher maturity, and it stayed almost constant when the maturity increased further (Emmanuel et al., 2016a). As a result of maturation, the proportion of aromatic compounds increased, and the weight percentage of aliphatic $C-H$ bonds decreased. These chemical alterations led to an increase in kerogen density, and Emmanuel et al. (2016a) concluded that as the elastic modulus of synthetic organic polymers are highly and directly correlated with density, and they expected the same behavior for kerogen. Eichmann et al. (2018) measured the modulus of OM-rich shale samples using the PeakForce QNM™ method. The samples had different thermal maturities from immature to overmature, but they did not observe a specific correlation between the maturity level and the elastic modulus of OM. Yang et al. (2017) showed that AFM-IR is capable of chemically mapping different types of OM within the OM-rich shales, but this method is not capable of measuring the mechanical properties quantitatively. Therefore, it is worthwhile to combine the stiffness measurement using AFM (PeakForce QNM™) with the AFM-IR to find the relationship between the types of OM and stiffness.

## 4. Comparison between nanoscale mechanical experimental methods

Overall, three nanoscale mechanical experimental methods have been applied to measure the mechanical properties of shales at nano-to micro-scales. Two of them are nanoindentation-based method (gridded-nanoindentation and modulus mapping) and the other one is an AFM-based method (PeakForce QNM™). Table 4 presents a summary of advantages and limitations of these three methods.

### 4.1. Comparison between the gridded nanoindentation and modulus mapping techniques

According to Ulm and Abousleiman (2006), the main assumption of the gridded-nanoindentation method is that the two component materials have a distinguishable difference in their mechanical properties. Therefore, for complex material such as shale that consists of more than two components, designing the pattern for gridded-nanoindentation is crucial to measure the moduli of the single component or phase (i.e., a distinct mineral), instead of measuring the moduli of the mixture of different minerals. However, the real contact force, the phase difference, and the amplitude of the tip are mapped directly in the whole area of interest in the modulus mapping method. Consequently, the reduced modulus is mapped in the whole area of interest. This capability of modulus mapping makes it possible to compare the recorded reduced modulus maps with the recorded SEM images of the same area.

For the complex materials such as shale, it is necessary to combine the gridded nanoindentation with statistical methods, such as deconvolution techniques or multivariate clustering techniques, to construct a histogram of each mechanical property over the nanoindentation area.

**Table 4**
The summary of advantages and disadvantages of each nanoscale mechanical characterization method.

| Method | Advantages | Limitations |
|---|---|---|
| Gridded Nanoindentation | Compatible for measuring both elastic and plastic properties | Finding the accurate indentation depth and spacing of the pattern for gridded-nanoindentation which are crucial to measure the moduli of single component (i.e., a distinct mineral) |
| Modulus Mapping | Measuring the contact force, reduced modulus, the tip's amplitude, and the phase difference in the form of two dimensional (2D) images, with these 2D images easily comparable with the EDS elemental maps and SEM images. | The modulus mapping based on an assumption that the indenter tip has a specific shape such as a sphere, and the measured storage modulus is highly affected by the radius of contact. |
| PeakForce QNM™ (AFM-based) | Measuring the reduced modulus continuously on the surface of sample, with the measured data easily comparable with the EDS elemental maps and SEM images. | Applying a low force to measure the elasticity modulus makes it hard to measure the creep properties. |

The mean and standard deviation of the measured indentation modulus and hardness for each mechanical phase are crucial statistical parameters. Moreover, the probability that one set of recorded measurements (indentation modulus and hardness) belongs to a specific mechanical phase is the main basis for the algorithm to determine different mechanical phase. Therefore, the level of uncertainty for the measured indentation modulus and hardness for each mechanical phase are important in the accuracy of combined gridded nanoindentation with statistical methods.

In relation to investigating the creep behavior, a quasi-static nanoindentation creep test can provide modulus and hardness of the sample at the end of the creep test (maximum indentation depth), but it is important to understand how a sample behaves during the test. In the modulus mapping based on the nano-DMA method, the average value of the quasi-static force is kept constant, but it fluctuates a little because of the need to measure the contact stiffness continuously. The modulus is calculated at the start of the holding stage, and this is used to measure the contact area continuously. This allows an accurate and continuous calculation of the average penetration depth and contact pressure over time. Therefore, modulus mapping based on the nano-DMA method is capable of investigating the creep behavior during the test, unlike gridded nanoindentation.

The contact area is calculated differently in nanoindentation and modulus mapping. According to Oliver and Pharr (2004), the projected contact area in nanoindentation is determined from the area function (Eq. 23 in Oliver and Pharr, 2004). However, the modulus mapping is based on an assumption that the indenter tip has a spherical shape. Therefore, the contact area in the modulus mapping has a spherical shape, and the storage modulus is highly affected by the radius of contact. This difference in the way contact area is measured causes a difference in the accuracy between these two methods, and the nanoindentation method is more accurate for modulus measurement than modulus mapping (Zhao et al., 2019). The relationship between the size of the contact area and the resolution of nanoindentation and modulus mapping was studied by Zhao et al. (2019). In the modulus mapping method, the measured contact radius for organic matter and the stiff minerals is approximately 40 nm and 10 nm, respectively, and the estimated contact areas are smaller than $7.85 \times 10^3 nm^2$ (Zhao et al., 2019). A nanoindentation test was conducted on quartz in the same test area that modulus mapping was conducted. Zhao et al. (2019) measured the contact area of indentation using the area function and estimated it to be $3.75 \times 10^5 nm^2$. The contact area of the nanoindentation method on the softer constituents is most likely larger than for quartz. Therefore, the measured reduced modulus in the modulus mapping method has a higher lateral resolution compared to measured indentation modulus in the nanoindentation method.

### 4.2. Comparison between the gridded nanoindentation and AFM-based techniques

The PeakForce QNM™ method has advantages and disadvantages compared to the gridded nanoindentation method. The main advantage of this method is that it provides a continuous measurement of the modulus for the different mechanical phases on the surface of shale samples, whereas gridded nanoindentation is based on a grid of nanoindentations over the specific area of the shale sample (i.e., the area of interest). Each indentation is made with a specific spatial distance from the other indentations. It is possible to compare the resulting modulus map from AFM-based methods (PeakForce QNM™ and other similar methods, explained at the start of Section 3.1) with the EDS elemental maps. This comparison is straightforward and can be used to determine the mineralogy of each mechanical phase (Eliyahu et al., 2015). The results from gridded nanoindentation and deconvolution methods are also comparable to the EDS elemental maps, but this comparison is not straightforward because the final product of gridded nanoindentation techniques is not a modulus map. The resulting deformation on the sample surface in the PeakForce QNM™ method is approximately 5–20 nm (Eliyahu et al., 2015; Emmanuel et al., 2016a; Li et al., 2018a), but nanoindentation-based methods induce deformations usually at the scale of hundreds of nanometers, and in some cases up to micrometers (Ulm and Abousleiman, 2006; Deirieh et al., 2012). PeakForce QNM™ is a non-destructive method because a relatively low force is applied to the sample surface (Li et al., 2018a; Zeng et al., 2018). It is worth mentioning that other kinds of the AFM force mapping methods which apply a relatively low force (explained at the beginning of Section 3.1) are also non-destructive.

### 4.3. Comparison between modulus mapping and AFM-based techniques

Both the modulus mapping based on Nano-DMA and second mode of AFM force mapping methods, such as PeakForce QNM™, provide a modulus map of the sample surface, but there is a difference between the type of modulus that these two methods are measuring. The nano-DMA method applies a higher force, and as a result it measures the viscoelastic behavior of the sample. However, the methods based on the second mode of AFM force mapping apply a lower force and measure the elasticity modulus (Young's modulus) of the sample. The nano-DMA and gridded nanoindentation methods can be utilized to measure the creep behavior of the shales at the nanoscale (Liu et al., 2018a; Slim et al., 2019), while it is not easy to apply the second mode of AFM force mapping methods to investigate creep behavior.

## 5. Conclusions and recommendations

The geomechanical characterization of shales is of great importance for different industrial practices such as carbon sequestration, hydraulic fracturing of shales, wastewater injection, and coal bed methane production. This review discusses current understanding of the major mechanical properties of OM rich shales with respect to their chemical properties such as mineralogy, TOC content and maturity level. Important mechanical properties include stiffness (Young's modulus and Poisson's ratio), mechanical strength under different loading conditions, and fracture toughness. The other crucial aspect of geomechanical characterization is the experimental conditions such as temperature, or if the experiments were conducted under the original stress condition. Nanoscale to microscale experiments on shale samples require a much smaller sample as compared to the macroscale experiments, and this is a huge advantage because it is possible to use drill cuttings for nanoindentation or AFM experiments. One main shortcoming of nanoindentation and AFM approaches is that both methods measure the mechanical properties very near to the surface of sample, but the constituents of OM-rich shales (both minerals and OM) have small sizes in micrometer and nanometer scales. Therefore, nanoindentation and AFM are useful to measure the mechanical properties of OM-rich shales.

Both the nanoindentation based and AFM based methods classify OM rich shales into 4 mechanical phases (each mechanical phase is the group of minerals/components that have the same mechanical property such as Young's modulus): Group 1) the softest components (Young's modulus of 0–25 GPa), group 2) clay minerals with mean value of Young's modulus ~29 GPa, group 3) isolated grains, which are mainly carbonate minerals, with average Young's modulus of ~58 GPa, and group 4) stiff minerals (mainly quartz, feldspars, and pyrite) with an average Young's modulus of higher than 90 GPa. The nanoscale mechanical characterization of OM-rich shales shows that the OM is the most pliable compound. Moreover, it is impossible to apply traditional rock mechanics experiments (Uniaxial Compressive Strength test, Triaxial Compressive Strength test, etc.) on the OM which is embedded within shales. Therefore, most of these nano-microscale experiments were focused on OM.

Based on the previously published research and considering the limitations of nanoindentation-based and AFM-based mechanical

characterization methods, the following recommendations are made for improving the mechanical characterization of OM rich shales:

1) As kerogen is the main component of OM, it is possible to measure the fracture toughness of kerogen using nanoindentation. Measuring the fracture toughness of kerogen can bring more insight on the possibility of fracture initiation within the OM.
2) It is important to investigate if different types of kerogens have different creep behavior, or the relationship between the maturity and creep behavior of OM needs to be investigated.
3) It is still unclear whether higher maturation results in the higher strength of OM or not. Moreover, the relationship between the impact of thermal maturation on the chemical structure of OM and how this relationship can impact the strength of OM needs to be investigated.
4) As the frictional properties of the fracture interface are highly important for understanding the interaction of natural fractures and hydraulic fractures, it is highly important to investigate the frictional properties of OM (how does friction change under different sliding velocities, normal forces, etc.).

## Author statement

The manuscript is approved by all authors, who have been involved in the whole process without a conflict of interest. F. Sobhbidari conducted the work and prepared the 1st draft, and Q. Hu offered the original concept, editing and financial support.

We would also like to declare that the work in this manuscript is an original research that has not been published previously or submitted to another journal.

## Declaration of competing interest

The manuscript is approved by all authors, who have been involved in the whole process without a conflict of interest. We would also like to declare that the work in this manuscript is an original research that has not been published previously or submitted to another journal.

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